- If $|V| > 3$ and G doesn't have any cycle with length 3, then $|E| \leq 2|V| - 4$.

So, as a first step in a planarity test algorithm, we can check, in linear time $O(V+E)$, both conditions[10] above: if either doesn't hold, we already know the answer is "non-planar".

There are several algorithms to test for planarity; while none of them is particularly easy to implement, many are also inefficient: the first efficient algorithm, running in worst-case linear time, was derived only in 1974 by Hopcroft and Tarjan.

The inefficient algorithms that had been develop before, instead, would take up to $O(|V|^3)$, or even worse, as we'll see.

One way to try and improve the situation is using the divide-and-conquer strategy to break down the original graphs into smaller subgraphs that can be tested separately.

This is possible thanks to the following two lemmas:

1. A graph is planar if and only if all its connected components are planar;
2. A graph is planar if and only if all its biconnected components are planar.

In chapter 14 we have already given the definition of **connected graph**: G is connected when from any vertex $v \in G$ it is possible to find a path to any other vertex $u \in G$. If a graph is not connected, we can define its connected components as the maximal disjoint subgraphs of G that are connected.